

Class 2:

1. Find by double integration, the area lying between the parabola $y = 4x - x^2$ and the line $y = x$
ans: 4.5
2. Find the volume bounded by the xy-plane, the cylinder $x^2 + y^2 = 1$ and the plane $x + y + z = 3$
ans: 3π
3. Find the average value of the function e^{x+y} over the region $R = [0,2] \times [0,2]$ ans: $\frac{(e^2 - 1)^2}{4}$
